



Lab 03: DNS & DHCP Servers

DNS · DHCP · Google · Facebook Servers · Switch · PC0 · PC1 · PC2

Lab #03

Topic: Network Services

Tool: Cisco Packet Tracer

Cert: CompTIA Network+

1. Lab Objectives

- Configure DHCP server to automatically assign IPs to PC0, PC1 and PC2.
- Configure DNS server to resolve google.lab and facebook.lab to IP addresses.
- Set up simulated Google and Facebook web servers with static IPs.
- Verify DHCP lease acquisition and DNS name resolution from all three PCs.
- Relate to Network+ objective 1.6: DNS and DHCP services.

2. Network Topology

Switch0 (2960) is the central hub. DNS server on Fa0/5, DHCP server on Fa0/1, Google server on Fa0/6, Facebook server on Fa0/7. PC0 on Fa0/2, PC1 on Fa0/3, PC2 on Fa0/4. All share the 192.168.1.0/24 subnet.

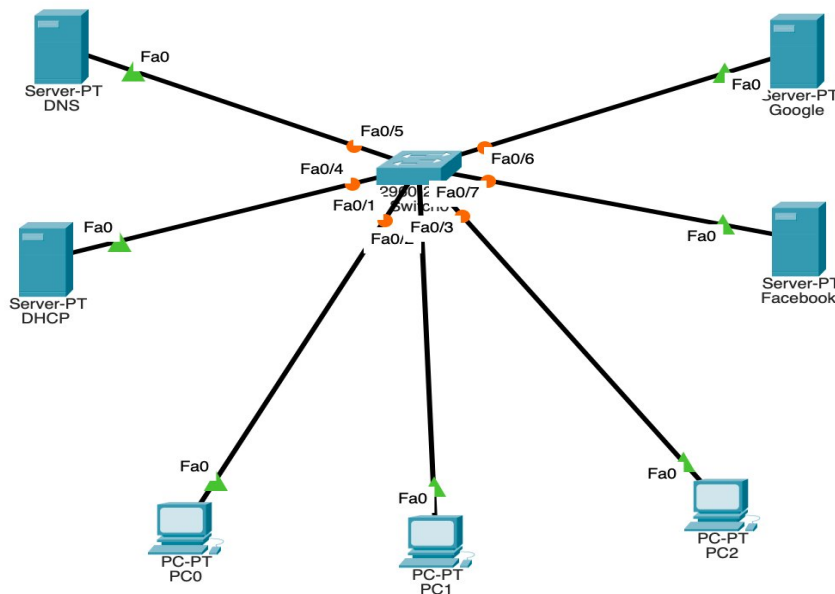


Figure 1 – Star topology: 4 servers + 3 client PCs all connected to Switch0

3. IP Addressing Table

Device	Role	Switch Port	IP Address	Subnet Mask	Notes
DNS	DNS Server	Fa0/5	192.168.1.2	255.255.255.0	Static



DHCP	DHCP Server	Fa0/1	192.168.1.3	255.255.255.0	Static
Google	Web Server	Fa0/6	192.168.1.10	255.255.255.0	Simulated google.lab
Facebook	Web Server	Fa0/7	192.168.1.20	255.255.255.0	Simulated facebook.lab
PC0	Client	Fa0/2	DHCP (192.168.1.50-255.255.255.0)	255.255.255.0	Auto-assigned
PC1	Client	Fa0/3	DHCP (192.168.1.51-255.255.255.0)	255.255.255.0	Auto-assigned
PC2	Client	Fa0/4	DHCP (192.168.1.52-255.255.255.0)	255.255.255.0	Auto-assigned

4. Configuration

4.1 DHCP Server

Pool Name	LAN_POOL
Default Gateway	192.168.1.1
DNS Server	192.168.1.2
Start IP	192.168.1.50
Subnet Mask	255.255.255.0
Max Users	50

4.2 DNS A Records

Hostname	Record Type	IP Address
google.lab	A Record	192.168.1.10
facebook.lab	A Record	192.168.1.20
dhcp.lab	A Record	192.168.1.3

4.3 Enable DHCP on PCs

PC0 / PC1 / PC2 > Desktop > IP Configuration > select: DHCP

Wait 2-3 seconds – IP, subnet mask, gateway and DNS auto-assigned

Expected lease example for PC0:

IP Address : 192.168.1.50

Subnet Mask: 255.255.255.0

Gateway : 192.168.1.1

DNS Server : 192.168.1.2

5. Verification

PC0>ipconfig ! Confirm DHCP lease received



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PC0>ping 192.168.1.2      ! Ping DNS server by IP
PC0>ping google.lab       ! Test DNS name resolution
PC0>ping facebook.lab     ! Test second DNS A record
PC0>ping 192.168.1.51     ! Ping PC1 (also a DHCP client)
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6. Key Takeaways

- DHCP uses the 4-step DORA process: Discover, Offer, Request, Acknowledge.
- DNS A records map hostnames to IPv4 addresses — the most common DNS record type.
- DHCP option 6 (DNS server) and option 3 (default gateway) are sent automatically with the lease.
- Simulating real services (Google, Facebook) builds practical DNS troubleshooting skills.
- Servers always use static IPs — DHCP auto-assignment is for client devices only.